Using AI to design retrocomputer PCBs challenge

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lyn...@yahoo.com

1:00 PM (9 hours ago)

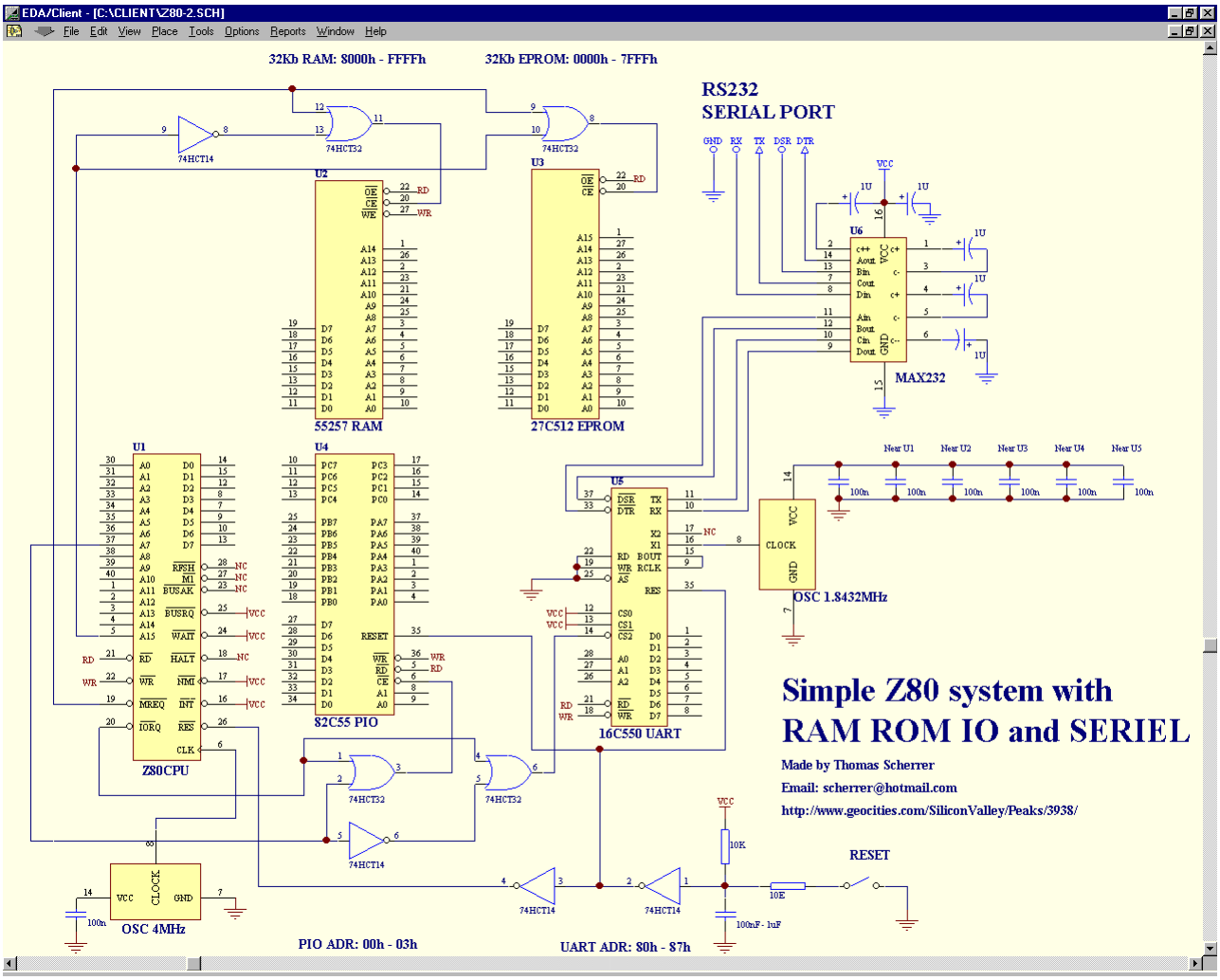
to retro-comp

Hi

Of late I've been interested in what can be done with AI to design retrocomputer PCBs. I've tried various things with mixed results. My best results so far are with Grok.

I decided to make a challenge to use AI to design a fully working PCB for a retrocomputer using this classic system on Z80.info

http://www.z80.info/gfx/z80test.gif



I chose this system because it was the first system I built using a breadboard about 20 some years ago. Since it's been around for a long time, it seemed like a good reference system. Also, I am familiar with it and have implemented in KiCAD manually so I can compare it to something.

Using Grok, I uploaded the GIF and the following prompts, it generated a KiCAD project, a schematic, and a PCB including a simple monitor program. Unfortunately, the schematic and the PCB layout have errors which prevent them from loading in KiCAD so the results clearly still need some work.

> *create Z80 simple system in KiCAD including schematics and PCB layout with traces*
>
> *make the files available for download and also generate an updated design with*
> *1. more complete fully-wired schematic*
> *2. double-sided version*
> *3. a CSV BOM*
> *4. a simple monitor ROM with Z80 assembler source code listing for TASM*
>
> *create the remaining standard library symbols (62256 , 27C512 , 8255 , 16C550 / PC16550D , MAX232 , 74HCT14 , 74HCT32) with each symbol in a separate library*
>
> *Create detailed images of schematic (without overlay of networks or net labels) and final PCB layout*
>
> *expand monitor ROM features and revise schematic for clarity*

It would be great if AI were capable enough that we could load a schematic image and/or image of a PCB and the AI could design a fully working retrocomputer with just a few prompts. That's probably beyond the state of the art at the moment although it is clearly improving.

Have you used AI to produce working retrocomputer PCBs? How did you do it?

Feel free to give this a try and post your results



7alken 3:00 PM (7 hours ago)

to retro-comp

hi Andrew, few days ago I noticed on X some link to this repo https://github.com/John-owo/pcb-agent-workflow which seemed related to Kicad and AI PCB design (in fact also manufacturing, not interested much), I didnt noticed he has in fact apart from traditional chinese also english tab in readme, but was curious what all this project covers, so I asked codex CLI to fetch it and translate (so, my excuses, inflation... feel free to see into original also) and try to describe for what is used claude here and if is possible to review this and modify it to codex-only, AND try to incorporate int that also testing of java FreeRouting, which is single automated thing I tested for while till now, only scratched surface though, as the settings of that was manually confusing for me ... so I expect the machine will be able to steer this better, but original project probably uses KiCad directly using APIs ... I really wanted to understand how this all works, so I made that my repo codex-only (and FreeRouting) towards my proposed boards (without any specific info, though ...) ... it did something during about a hour, but I had not more time and focus here for the results, only I was reading all the generated and converted docs in my preffered ascidoctorjs css/js format, I feel its really faster for me ... it is NOT TESTED, absolutelly ... if somebody wants to pickup and continue ... I am not able to focus more now - but it seems to be interesting, all the terms, glossary, process, details even up to manufacturing...

docs in my repo are the worklog "in my sense", we moved original docs into guides, maintained them and created ours too ... its really only material to study the thing more;
(again, apologies that I missed original repo already IS partially in english too ... and sure, browser translate works also)

https://github.com/apws/261029-codex-kicad-pcb

Petr



michelle...@gmail.com 3:01 PM (7 hours ago)

to retro-comp

Very interesting. I probably will never use AI for board design, because I enjoy the process of using KiCAD to design and layout the boards. Of course the auto-route plugin is in a way an AI thing. Interesting though, I watched a Dave Plummer YT video on him designing and building a 4MB RAM board for his PDP-11 using AI: "
I Made My Own RAM - But Will It Boot?" - https://www.youtube.com/watch?v=RG9O43-ueCU



lyn...@yahoo.com 7:51 PM (2 hours ago)

to retro-comp

Hi,
I've revised the AI prompts and posted them here along with the KiCAD project, schematic, and PCB layout file. See attachments

They didn't load for me in KiCAD when I tried them yesterday.

If anyone can successfully load these files or knows how to prompt Grok to make correct ones, please post.